RYERSON POLYTECHNIC UNIVERSITY

DEPARTMENT OF MATHEMATICS, PHYSICS & COMPUTER SCIENCE

PCS 125 FINAL EXAMINATION

DURATION: 2 hours **DATE: 15 December 2000**

Fill in the rectangles below and sign your name.

Section	Student Number	PRINT first name	PRINT last name

Signature: _____

EXAMINATION INSTRUCTIONS

1. This is a closed book examination.
2. A formula sheet (printed on both sides) is provided at the end of this exam and may be detached.
3. Non-programmable calculators are permitted.
4. Do all your calculations in this booklet. If you continue a problem elsewhere in this booklet, clearly direct the marker to your solution.
5. Problems 1, 6, and 7 are multiple choice.
6. Do all seven questions.
6. Please show all your work. The correct answer alone may be worth nothing.
8. The exam totals 50 marks and is worth 50% of your course grade.
9. Circle your professor's name below.

* M. Buckby
* M. Kolios
* A. Pejović-Milić
* W. Que
* B. Todorow
* B. Whelan

Question	Grade
1	_____
2	_____
3	_____
4	_____
5	_____
6	_____
7	_____

pg. 402

Marks

1. **Please select the nearest value to the correct answer from the choices given. Show your calculations and enter the letter of your choice into the box at the right of each question. Only the boxes will be marked.**

A horizontal travelling wave is generated in a string that is driven at one end at a frequency of 5.0 Hz. The amplitude of the wave is 12.0 cm and the speed of the wave is 20.0 m/s. The wave is travelling in the positive x-direction.

[1] a) Determine the angular frequency (in appropriate S. I. units) for this wave.
A) 15.7 rad/s B) 1.57 rad/s C) 31.4 rad/s D) 31.4s E) 3.14 s

[1] b) Determine the wave number (in appropriate S.I. units).
A) 0.64 m B) 0.64 m^{-1} C) 1.57 m^{-1}. D) 1.57 m E) 15.7 m^{-1}

[1] c) The linear density of the string is 0.020 kg/m. What is the tension in the string?
A) 0.4 N B) 1×10^{-4} N C) 2000 N D) 8 N E) 4 N

[1] d) Determine the power transmitted to the wave.
A) 0.75 W B) 1.51 J C) 2.84 W D) 2.84 J E) 1.51 W

[1] e) Using the values obtained in 1a) and 1b) above, write the equation in S.I. units that describes the vertical displacement of a point on the wave.

A) $y = 0.12 \sin(1.57x + 31.4t)$
B) $y = 0.12 \sin(0.64x - 15.7t)$
C) $y = 0.12 \sin(3.14x + 1.57t)$
D) $y = 0.12 \sin(1.57x - 31.4t)$
E) none of these choices

[1] f) In the equation of part 1e), the S.I. units for (y, x, t) are, respectively,
A) (m, m^{-1}, s) B) (m^{-1}, m, s) C) (m, m, s) D) (m, m^{-1}, s^{-1}) E) (m^{-1}, m^{-1}, s)

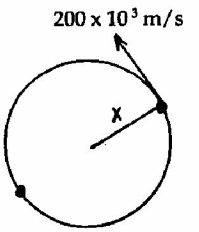
Marks

2. You are standing on the edge of a lake and a pier is being built on the opposite edge of the lake. When some blasting occurs, the sound travelling through the water arrives 3.0 seconds before it reaches you through the air.

[6] Use v_{sound} (in air) = 343 m/s and v_{sound} (in water) = 1493 m/s to determine the distance, d, across the lake (in km).

3. Two stars revolve in a circular orbit about a center of gravity that is midway between them. The masses of the stars are equal. The orbital speed of each star is 200×10^{3} m/s and the orbital period of each is 9.0×10^{5} s.

[2] a) Determine the orbital radius, x (in meters).



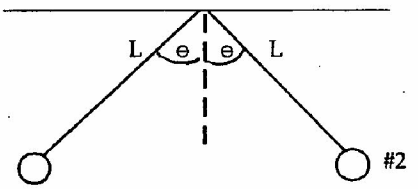
[2] b) Determine the mass of each star (in kg).

Hint: centripetal force = $F = \frac{mv^2}{x}$, where x = radius of circle.

[2] c) What is the minimum additional speed (in m/s) that must be imparted to one of the stars if it is to escape from orbit?

Marks

4. Two identical point charges, each having a mass $m = 0.1$ kg and a charge, q, hang from two massless strings as in the figure. Each string has a length of $L = 50.0$ cm and makes an angle of $\theta = 40°$ with respect to the dotted central line. Determine:



[2] a) the equation for the x-components of the forces (electrostatic, gravitational, etc.) acting on mass #2;

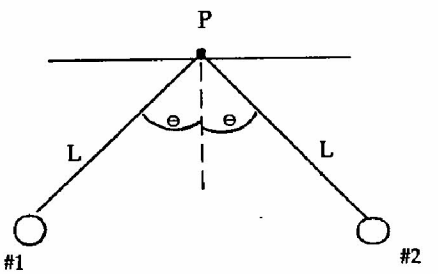
[2] b) the equation for the y-components of the forces (electrostatic, gravitational, etc.) acting on mass #2;

[2] c) the magnitude of the electrostatic <u>force</u> acting on mass #2;

[2] d) the magnitude of the tension in the right string.

Marks

5. Two identical point charges, each having a mass m = 0.1 kg and a charge q = 6.0 μ C, hang from two massless strings as in the figure. Each string has a length of L = 50.0 cm and each string makes an angle $\theta = 40°$ with respect to the dotted central line.



[1] a) Please circle the correct choice:
The magnitude of the electric field at point P is 3.31×10^5 V/m.
The direction of the electric field at point P is along the:

positive x-axis positive y-axis positive z-axis

negative x-axis negative y-axis negative z-axis

[3] b) Calculate the net electric potential at point P.

[2] c) A 5.0 μC charge is now placed at point P . What is the magnitude of the electrostatic force exerted upon this 5.0 μC charge?

[2] d) Fill in the blanks : One electronvolt (ev) is defined as the _____ acquired by an electron or proton as it moves through a potential difference of _____.

Marks

6. **Please select the <u>nearest value</u> to the correct answer from the choices given. Show your calculations and enter the <u>letter</u> of your choice into the box at the right of each question. <u>Only</u> the boxes will be marked.**

Deuterium ions, each consisting of one neutron and one proton and having a mass of 3.32×10^{-27} kg, are accelerated from rest through a potential difference of 60.0 kV.

The ions then enter a velocity selector in which the electric field intensity is 4.0 kV/m. Ions having the correct velocity exit into a uniform magnetic field that has the same magnitude and direction as the magnetic field in the velocity selector.

a) Determine the speed of the ions in S.I. units.
1] A) 1.70×10^{6} B) 3.39×10^{6} C) 2.40×10^{6} D) 5.78×10^{12} E) none of these

b) The answer selected in part 6a) has S.I. units of :
[1] A) m/s B) km/s C) km/hour D) cm/s E) none of these

c) Determine the magnitude of the magnetic field in S.I. units.
[1] A) 600 B) 1.67×10^{-3} C) 2.35×10^{-3} D) 1.18×10^{-3} E) none of these.

d) The answer selected in part 6c) has S.I. units of :
[1] [1] A) W B) T C) G D) N/C E) none of these

e) Determine the radius of the ions' orbit while in the magnetic field.
[1] [1] A) 14.9 B) 29.8 C) 21.1 D) 42.1 E) none of these

f) The answer selected in part 6e) has S.I. units of
[1] A) km B) mm C) nm D) m E) none of these

Marks

7. **Please select the value that is closest to the correct anser from the choices given. Show your calculations (if applicable) and enter the letter of your choice into the box at the right of each question. Only the boxes will be marked.**

[10]

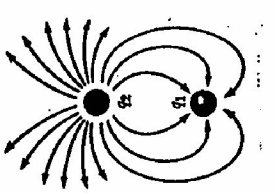
a) A cubic zirconia prism has an apex angle of 30 °. When the prism is entirely surrounded by air, its angle of minimum deviation is 39.4 °. The index of refraction for cubic zirconia is: A) 1.85 B) 2.20 C) 3.60 D) 0.28 E) none of these.

b) A particle at rest has a total positive charge of 25.0 μC. Through what potential difference must this charged particle be accelerated in order to increase its potential energy by 50.0 mJ?
A) 2.2V B) 50 mV C) 2000V D) 0.5V E) none of these.

c) A nucleus has a charge of 1.472×10^{-17} C. How many protons are in this nucleus?
A) 108 B) 113 C) 9×10^{9} D) 92 E) none of these.

d) The figure shows the electric field lines for two close point charges. The ratio $\frac{q1}{q2}$ is

A) 2 B) 3 C) 1/2 D) 1/3 E) none of these.



e) The mass of Mars is 0.107 M_E and its radius is 0.53 R_E. The gravitational field g (in m/s^2) at the surface of Mars is approximately :

A) 2.63 B) 0.27 C) 3.73 D) 2.5×10^{-10} E) none of these

Marks

f) A standing wave is established in a 120 cm length of string which is fixed at both ends. When driven at 120 Hz, the string displays four loops (i.e., two complete waves). The fundamental frequency (in Hz) that will produce one loop is:

A) 480 Hz B) 30 Hz C) 60 Hz D) 120 Hz E) none of these

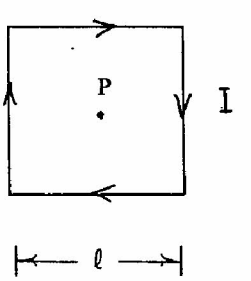
g) A Young's double slit experiment is performed with light from an Argon laser. The separation between the slits is 0.5 mm and the interference pattern appears on a screen that is 3.3 m away. The 2nd order maximum appears at 6.8 mm from the center of the pattern. The wavelength (in nm) is approximately:

A) 515 B) 5150 C) 1030 D) 650 E) none of these

h) An oil film (n=1.45) floating on water (n=1.33) is illuminated by white light at normal incidence. The dominant observed color in the reflected light is blue. The dominant color of the transmitted light is:

A) blue B) orange C) green D) purple E) none of these

i) A particle on a spring is pulled downward 4 cm from its equilibrium position and released. The maximum upward acceleration of the particle is 0.30 m/s^2. The maximum velocity (in m/s) is:

A) 1.10 B) 0.30 C) 0.11 D) 2.74 E) none of these

j) A conductor carrying a current I = 10.0 A is formed into a square. Each side is length ℓ = 0.5 m. The magnitude of the magnetic field at the center of the square (point P) is: A) 2.57 G B) 2.57×10^{-4} T C) 6.42×10^{-5} G D) 22.7×10^{-6} T E) none of these